AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T316
Advanced Networking

Midterm Examination (MTA)
FALL Semester 2025/2026

Date: 15 /Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and** ID on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: True/False Questions (10 marks)

Question 1: Put True/False for each of the following definitions (1 mark for each question)

1. Quality of Experience (QoE) primarily refers to low-level network metrics such as throughput and packet loss. (.....)
2. Asymmetric Digital Subscriber Line (ADSL) is a technology used to convert a traditional copper telephone line into a high-speed digital link. (.....)
3. MPLS (Multiprotocol Label Switching) introduces a shim layer that operates logically between Layer 2 (Data Link) and Layer 3 (Network) of the OSI model. (.....)
4. The DiffServ QoS model utilizes DSCP bits, allowing for a theoretical maximum of 64 distinct classes of service. (.....)
5. Network traffic that is tolerant to significant variations in delay and jitter is commonly referred to as inelastic traffic. (.....)
6. Latency and jitter are metrics that do not affect the overall quality of service (QoS) in a network. (.....)
7. Traffic Engineering (TE) mechanisms are used to ensure that all packets belonging to a specific flow follow a predetermined, consistent route across the network backbone. (.....)
8. Quality of Service (QoS) is strictly concerned with high-level user perception, rather than low-level network parameters like bandwidth and delay. (.....)
9. The Border Gateway Protocol (BGP) is primarily classified as a distance-vector routing protocol. (.....)
10. Jitter refers to the fixed time delay experienced by all packets in a single data flow. (.....)

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PART 2: Short Answer Questions (20 Marks)

Question 1: (10 marks)

A) Explain three advantages that make it useful in modern telecommunications networks. (6 Marks)?

B) Using the SONET/SDH STM-1 basic frame structure, it shows that the data rate for OC-3 or STM-1 is 155,520 Mbps, the payload rate is 150,336 Mbps, and the overhead rate is 5.184 Mbps. (Step-by-step using mathematical calculations are mandatory) (4 Marks)

Question 2: (10 marks)

A) It has been said that when IPv6 tunnels through IPv4 routers, IPv6 treats the IPv4 tunnels as link-layer protocols. Do you agree with this statement? Why or why not? (2 marks)

B) BGP advertises a route by including the prefix and set of attributes. How does BGP use the AS-PATH and NEXT-HOP attributes? (2 marks)

C) Compare single mode and multimode fibers in terms of cost, usability, and diameter? (6 marks)

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PART 3: Long Answer Questions (30 Marks)

Question 1: (15 marks)

A) What are the advantages of the differentiated Services (DiffServ)? (Highlight 4 advantages) (4marks)

B) Describe the difference between an explicit route and a hop-by-hop route. Under what conditions does an explicit route between an ingress LSR and an egress LSR coincide with the hop-by-hop route? (5 marks)

C) Highlight the differences between EBGP and IBGP in terms of (where is it run, where is it used?)? (6 Marks)

Question 2: (15 marks)

A) The transmission of light through an optical fiber is subjected to optical effects, known as impairments. Explain **all** the several types of impairments. (7 marks)

B) What is the maximum core diameter for a fiber if it is to operate in single mode at a wavelength of 1550 nm if the N.A. is 0.12 and $V = 2.405$? (8 marks)

End of questions

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